

AAC Consumer Edition - WS2812 8x32 Panel Interface

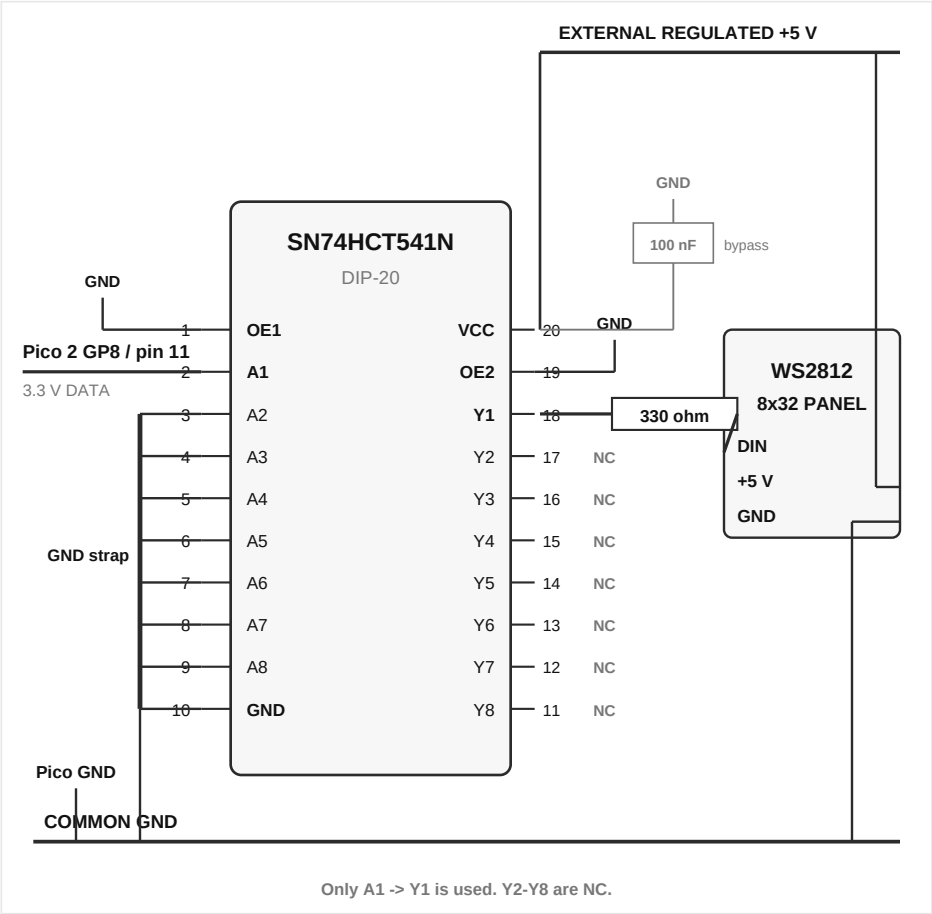
REVISION A | INTERNAL / PRIVATE - DIMENSION EIGHT LABS

1. PIN-TO-PIN WIRING TABLE

From	To	Connection / Notes
Pico 2 GP8 / physical pin 11	HCT541 pin 2 / A1	WS2812 DATA input to buffer
HCT541 pin 18 / Y1	330 ohm series resistor	Buffered DATA output
330 ohm series resistor	WS2812 panel DIN	DATA to panel
External regulated +5 V	HCT541 pin 20 / VCC	HCT541 supply
External regulated +5 V	WS2812 panel +5 V	Panel power
Common GND	Pico GND	Logic reference
Common GND	HCT541 pin 10 / GND	HCT541 ground
Common GND	HCT541 pin 1 / OE1	Active-low enable
Common GND	HCT541 pin 19 / OE2	Active-low enable
Common GND	HCT541 pins 3-9 / A2-A8	Unused inputs held LOW
Common GND	WS2812 panel GND	Panel return
100 nF ceramic capacitor	HCT541 pin 20 to pin 10	Local VCC bypass; place close to IC

2. HCT541 PIN USE AND VISUAL WIRING

Pin	Name	Connection
1	OE1	GND
2	A1	Pico 2 GP8
3	A2	GND
4	A3	GND
5	A4	GND
6	A5	GND
7	A6	GND
8	A7	GND
9	A8	GND
10	GND	GND
11	Y8	NC
12	Y7	NC
13	Y6	NC
14	Y5	NC
15	Y4	NC
16	Y3	NC
17	Y2	NC
18	Y1	330 ohm -> DIN
19	OE2	GND
20	VCC	External +5 V



3. BENCH NOTES

- WS2812 8x32 Panel is optional; GP8 belongs to this display only when the WS2812 driver is selected.
- HCT541 is used as a one-channel 3.3 V to 5 V non-inverting level shifter/buffer; only A1 -> Y1 is used.
- Unused inputs A2-A8 are tied LOW; unused outputs Y2-Y8 are left NC.
- Panel and HCT541 use a separate regulated 5 V supply during development; Pico and external supply grounds are common.
- Do not power the panel from Pico VBUS during initial bring-up. Pixel mapping/orientation will be established during bring-up.

WARNING: DO NOT CONNECT EXTERNAL +5 V TO A PICO GPIO PIN.
Verify +5 V polarity before connecting the panel; begin bring-up at low brightness/current.